



Creating a realistic sybil attack dataset for inter-vehicle communication

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Abstract

Sybil attacks, where malicious nodes use multiple identities, pose a critical threat to Vehicular Ad-Hoc Networks (VANETs). While Received Signal Strength Indicator (RSSI)-based detection offers a promising approach, existing research is hampered by the lack of realistic datasets that capture the complexities of real-world vehicular environments. Current datasets often simplify traffic conditions, neglecting the impact of congestion and packet drops on RSSI readings. This work addresses this gap by creating a large-scale, realistic VANET simulation environment for generating RSSI data. Using OpenStreetMap data for a region of Istanbul, we model realistic traffic flows calibrated with municipal data, incorporating packet loss due to congestion and interference. Critically, we employ a carefully calibrated log-normal shadowing model to capture the impact of environmental factors on RSSI. We implement four common Sybil attack methods with various power control strategies, creating a diverse set of attack scenarios. Rigorous validation against real-world traffic patterns, RSSI characteristics, attack distribution, and packet collisions demonstrates the realism of our dataset. We provide a sample dataset, along with our open-source simulation environment, enabling researchers to develop and evaluate robust Sybil attack detection mechanisms for real-world VANETs

Keywords Dataset · VANET · Received signal strength indicator (RSSI) · Sybil attack · Inter vehicle communication (IVC)

1 Introduction

Vehicle-to-everything (V2X) communication is crucial for Intelligent Transportation Systems (ITS), enabling numerous safety and efficiency applications. These applications rely on the exchange of Cooperative Awareness Messages (CAMs) [1] and Decentralized Environmental Notification Messages (DENMs) [2], which contain time-sensitive information. While pseudonyms enhance privacy [3], they also introduce security vulnerabilities, particularly Sybil attacks [4]. In a Sybil attack, a malicious node uses multiple identities to disrupt traffic flow, disseminate false information, or even endanger vehicles by transmitting misleading data.

Such attacks, whether intentional or due to sensor malfunction [5], can have severe consequences.

Securing V2X communication requires both proactive (e.g., PKI-based certificates [6]) and reactive measures [7]. Reactive security employs Misbehavior Detection Systems (MDS) [5, 8, 9] to identify malicious nodes. Among various attacks, position-related attacks, like Sybil and false position attacks, are prevalent. Leveraging physical signal characteristics, such as Received Signal Strength Indicator (RSSI), offers a cost-effective approach for message origin localization [10, 11]. However, the effectiveness of RSSI-based Sybil attack detection depends crucially on the availability of realistic datasets.

Real-world testing of V2X applications and MDSs is challenging. Simulation environments offer a viable alternative, but existing datasets often fall short in capturing the complexities of real-world traffic [12–14]. Critically, the impact of congestion and packet drops on RSSI readings is frequently neglected. In congested scenarios, signal propagation is significantly altered, leading to packet loss and

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impacting RSSI values. These factors are essential for developing robust detection mechanisms. For example, during congestion, RSSI values might fluctuate significantly due to increased interference and packet collisions. A Sybil attacker could exploit these fluctuations to blend in with legitimate vehicles, making detection more difficult. Existing datasets, by not modeling these effects, fail to prepare detection algorithms for these realistic challenges.

Several studies have highlighted the importance of realistic simulation environments for V2X security. For instance, the work in [15] emphasizes the need for accurate modeling of communication channels and network conditions. Furthermore, [16] discusses the challenges of simulating realistic traffic scenarios, including the impact of vehicle density and mobility. The study in [17] explores the use of machine learning for misbehavior detection in VANETs and highlights the importance of training and testing with representative data. Additionally, [18] investigates the impact of different attack models on the performance of intrusion detection systems in VANETs, underscoring the need for diverse and realistic attack scenarios in datasets.

While numerous MDS methods exist, the scarcity of publicly accessible datasets incorporating realistic traffic and RSSI characteristics, especially under congested conditions, hinders progress. Datasets like VeReMi [12], F2MD [13], and VeReMi-Extension [14] primarily focus on plausibility checks. Although some studies explore RSSI-based misbehavior detection [19–21] and power manipulation attacks [11], a dedicated dataset for evaluating these methods in realistic congestion scenarios is lacking.

This paper addresses this crucial gap by introducing a new, realistic VANET simulation environment and dataset specifically for Sybil attack detection. For this paper, the main contributions are as follows: (1) We simulate traffic flows in Istanbul, a city with notoriously heavy traffic, using OpenStreetMap data and calibrating traffic density based on real-world municipal data. This realistic congestion modeling allows evaluation under diverse congestion levels, a key improvement over existing datasets. (2) We use a calibrated log-normal shadowing model to capture environmental impacts, including congestion and packet drops, on RSSI readings. This realistic RSSI modeling is essential for generating realistic RSSI values and developing robust detection methods. (3) We implement various Sybil attack methods, including those where attackers manipulate transmission power, enabling the evaluation of a wider range of attack strategies.

The rest of the paper is organized as follows: Section 2 reviews related work. Section 3 details the simulation environment. Section 4 describes the Sybil attack scenarios. Section 5 presents dataset validation. Finally, Section 6 concludes the paper.

2 Related work

Sybil attacks pose a significant challenge to the security and integrity of VANETs [22]. Researchers and developers require realistic and comprehensive datasets to develop and rigorously evaluate robust detection methods. Existing sybil attack detection methods rely on a diversity of approaches, including packet inspection [13, 23] and the analysis of physical signal attributes [21, 24], with a strong emphasis on the RSSI. However, a major obstacle in evaluating and validating RSSI-based detection methods is the lack of publicly available datasets that specifically include realistic RSSI values and a variety of sybil attack scenarios. In this section, we will examine existing datasets both designed for misbehavior and sybil attack detection. Additionally, we will examine works that have modified existing datasets or evaluated new attack scenarios, considering their applicability to our simulation environment.

VeReMi [12] is the first public dataset for comparing and evaluating misbehavior detection methods in VANET. The VeReMi dataset is derived from the Luxembourg SUMO Traffic (LuST) scenario [25] and built within the Veins simulation environment [26]. It provides message logs, RSSI values, and includes diverse malicious message types created by multiple attack scenarios. The open-source nature of VeReMi allows the creation of custom datasets with new attack scenarios by modifying and rerunning the simulation. However, its default configuration limitations include low beaconing frequency (1Hz), small simulation area (3.6 km^2), low traffic density, and a lack of built-in sybil attack scenarios, making it unsuitable for evaluating RSSI-based sybil attack detection methods. The simulation uses the outdated Veins 3.0, and due to changes in the physical and medium access control layers, substantial updates would be required to make it compatible with the current version 5.2.

Kamel et al. [13] proposed F2MD, an open-source misbehavior detection simulation framework designed for developing, testing, and evaluating misbehavior detection algorithms. Built on Veins [26], F2MD offers a range of V2X attacks, detection algorithms, pseudonym change policies, visualization features, and performance metrics. It includes two traffic scenarios: LuST [25] and a Paris-Saclay area scenario [8], allowing simulations across customizable map sizes up to 150 km^2 . However, modifications are needed to add RSSI support and new types of sybil attacks [11] to evaluate RSSI-based sybil attack detection methods. Our work incorporates F2MD components, including Veins simulator integration, the application layer, message creation, and attack initialization.

The VeReMi Extension dataset [14] expands the VeReMi [12] by incorporating realistic sensor error models [8], new attacks, and additional information in CAMs. Generated

using the F2MD [13] simulation environment, the dataset includes multiple sybil attack scenarios. However, since it is generated using the F2MD framework, the VeReMi Extension inherits the F2MD's [13] limitations regarding RSSI support. Additionally, the dataset covers a small area (1.6 km^2) with low traffic density and low beaconing frequency (1 Hz), which makes it unsuitable for evaluating attack detection methods in dense and diverse traffic situations.

In [27], the authors proposed the DARE dataset designed for evaluating network wide misbehavior detection methods. It consists of reports of misbehaviors detected with plausibility checks by the vehicles in the F2MD [13] simulation environment. DARE shares the same attack models as the VeReMi Extension [14] but focuses exclusively on data compliant with the misbehavior reporting protocol [28]. Since it lacks simulation of RSSI measurements, DARE is unsuitable for evaluating RSSI-based sybil attack detection methods.

In their survey [22], the authors reviewed sybil attack detection methods, analyzed Cooperative Intelligent Transport Systems (C-ITS) standards, and identified relevant scenarios requiring further study. They also proposed a sybil attack detection dataset with 24 hours of data across highway and urban maps. However, the dataset's small map area (1 km^2) and lack of RSSI value simulation makes it unfit for evaluating RSSI-based sybil attack detection methods.

In [29], authors proposed the BurST-ADMA dataset designed for evaluating data-centric misbehavior detection methods. While they employed traffic scenario map and vehicle traffic generation techniques similar to the VeReMi dataset [12], BurST-ADMA diverges with its support for left-hand traffic, motorcycles, bicycles, and pedestrians. However, a significant limitation is that BurST-ADMA contains only faulty behavior attack types, making it only suitable for evaluating attack detection methods based on plausibility checks. Additionally, since the dataset is generated solely with a traffic simulator without a network simulator, it does not support network-level features like RSSI, delay, or packet loss.

Gonçalves et al. [30] proposed a dataset for misbehavior detection in VANETS. Their work focuses on DoS and fabrication attacks. The dataset provides 17 features derived from received CAM messages which basically contain, speed, heading, and acceleration. The dataset comprises seven scenarios with varying map sizes, ranging from the smallest at 0.07 km^2 to the largest at 1.47 km^2 and leverages both SUMO and NS3 simulators for generation. The dataset lacks RSSI or other physical measurements. Furthermore, power control schemes are not considered. Consequently, the dataset is unsuitable for evaluating sybil attack detection methods.

Yao et al. [19] introduced Voiceprint, a sybil attack detection method that analyzes RSSI time series of messages broadcast by neighboring vehicles. The method conceptualizes RSSI fluctuations as a vehicle's unique identifier, the

vehicle's voiceprint. By comparing voiceprints, vehicles can individually assess the legitimacy of their neighbors. However, Voiceprint's effectiveness relies on a high volume of RSSI samples, necessitating a beaconing rate of ten messages per second, the highest beaconing frequency according to standards [1, 31]. It's important to note that the evaluation of Voiceprint was conducted under ideal conditions, without considering packet losses that could occur due to collisions or Signal-to-Noise plus Interference Ratio (SNIR) limitations. The authors evaluated Voiceprint via a custom-built simulation environment consisting of a 2 km bi-directional highway. This work underscores the need for datasets supporting a beaconing rate of 10 Hz, as such datasets would enable the evaluation of a broader range of sybil attack detection methods that may rely on these frequencies.

Yao et al. [11] built upon their Voiceprint method [19] by evaluating its vulnerability to a new attack strategy. In this strategy, attackers manipulate their transmit power to evade detection, highlighting a limitation of Voiceprint's reliance on RSSI patterns. To address this attack, they proposed a time-series change-point detection method and adapted Voiceprint to utilize Service Channel (SCH) messages to reduce message observation time and false positives. While including SCH messages improves detection, they tested the method in a sparse simulation environment that implements transmission of 200 packets per vehicle in a 50 ms SCH period. Implementing such a high beaconing rate is challenging in both real-world scenarios and simulations due to potential channel congestion and computational overheads. Nevertheless, this work underscores the importance of including sybil attacks that employ transmission power manipulation in datasets. This will allow for the development and evaluation of robust detection methods against these sophisticated attack strategies.

Many researchers utilize existing datasets, including VeReMi [12] and its extension [14], along with the F2MD framework [13], for machine learning-based misbehavior detection. These resources are applied to tasks such as falsification detection [32, 33], anomaly detection [34], sybil attack detection [35], and multi-type attack detection [23]. However, several notable works demonstrate the advantages of modifying datasets to tailor them for specific research needs. Ercan et al. [24, 36] worked with a modified VeReMi [12] dataset to detect position falsification attacks. They included two new features: AoA and estimated RSSI distance. However, a critical limitation is that the AoA calculations rely on exact vehicle positions from the dataset rather than simulating signal reception. This leads to inaccurate reflection of real-world conditions where signal reception and AoA estimations can be affected by various factors. Chen et al. [37] enhanced the F2MD [13] by incorporating vehicle loop detector data for their sybil attack detection method. They added statistics such as the number of vehicles passing

and average vehicle speed on road segments. While loop detectors and AoA values can be valuable for misbehavior detection, our current focus is on a real-world like simulation environment. Therefore, we have not included these features [24, 37], as they necessitate significant infrastructure additions and specialized onboard hardware, which are not yet standard in VANET.

The simulations and datasets (Table 1) used in the testing stages of the sybil attack detection methods that we examined in this section do not adequately reflect large cities in terms of the map area and traffic density. In order to test RSSI-based methods in detail, simulation environments and data sets need to reflect interference effects on signal properties and packet losses due to vehicle density or environmental factors. The area selected for simulation should have various environmental factors affecting the radio signals, such as buildings, parks, and open spaces. For this reason, the selected area should be large enough or consist of multiple regions to increase the diversity of environmental factors. RSSI calculations used in the simulation should be adjusted to include these environmental factors. Moreover, newly created datasets should include sybil attack types that use power control methods.

3 System model

This section describes the scope of our simulation environment as well as its requirements to produce realistic results. Initially, the communication model of the simulation setup is explained, then map and traffic model aspects are discussed. Lastly, details of attacker models included in our simulations are given.

Our study aims to create an environment in which both sybil attacks and attack detection methods can be developed and evaluated. This requires a more realistic environment in which includes high traffic densities and more sophisticated sybil attacks. High-density traffic situations lead to packet losses, which are generally neglected by existing [13, 14] sybil attack detection methods. On the other hand, sophisticated sybil attacks that employ transmission power control [11] can survive against simple detection methods. Thus, it is important to include these properties in the simulation environment to develop and test more robust detection algorithms.

Our simulation environment includes the F2MD [13] application layer. The F2MD application layer provides pseudonym change policies. Due to the lack of correlation between consecutively used pseudonyms, surrounding vehicles face challenges in using local detection methods to identify sybil attacks. In our simulations, we focus on sybil attack detection on a local basis. Thus, we simply disabled the pseudonym change policies. However, for developing a global misbehavior detection system, it is possible to include

pseudonym change policies with a simple modification on the application layer.

3.1 Communication model in the simulation environment

The previous work on the detection of sybil attacks is carried out through inspection of the information in the CAMs [13, 23], the signal properties obtained while receiving the CAMs [10, 21], and comparing information extracted from these methods with environmental data [8]. We designed our dataset assuming that sybil attacks can be detected using any combination of this information. Our dataset consists of the CAMs sent during simulation, the logs of the vehicles receiving these messages, the physical properties of the received messages, and environmental information.

Our simulation environment covers the scenario where vehicles communicate directly over Dedicated Short Range Communications (DSRC) without the support of infrastructure components. Vehicles periodically broadcast CAMs 10 times per second in accordance with both European Telecommunications Standards Institute (ETSI) [1] and The Society of Automotive Engineers (SAE) [31] standards. CAMs contain the vehicle's identity, location, velocity, acceleration, and direction. All vehicles within the range supported by DSRC could receive these messages and measure the RSSI values. A simple sybil attack scenario in which an attacker impersonates other vehicles and tries to direct nearby vehicles to alternate routes is shown in Fig. 1.

3.2 Map and traffic model

Development and testing of sybil attack detection methods based on the physical features of signals require measurement of signal properties in an environment that reflects real-world conditions. Therefore, the simulation environment should include the factors that affect the signal properties. For example, communication between two vehicles in urban areas can vary considerably depending on whether the vehicles are on the same street or on different streets with buildings in between. For this reason, features such as open spaces, parks, forests, buildings, and farms should be labeled and used in the simulation. Section 4.1 provides a detailed description of the map generation process for our dataset.

Another factor affecting radio signals is the number of broadcasting vehicles. The smaller number of vehicles involved in the communication minimizes collision-related packet losses. Thus, it is easier to detect attacks using the signal features in these cases. However, in cases of dense traffic, the number of communicating vehicles may exceed the channel capacity. Therefore, a dataset used to detect sybil attacks must contain packet loss cases. For this reason, we designed our simulation environment to cover heavy traffic

Table 1 Comparison of datasets available for misbehavior and sybil attack detection

	VeReMi [12]	F2MD [13]	VeReMi Ext. [14]	DARE [27]	[22]	BurST ADMA [29]	[30]	[19]	[21]	Our dataset
Publicly available	YES	YES	YES	YES	YES	YES	YES	NO	NO	YES
# vehicles	48,015	Adj.	30,413	82,925	86,747	730	176	20 - 200	1000 - 10000	7,995
# genuine veh.	38,344	Adj.	21,289	61,607	62,421	666	94 - 166	Unknown	950 - 9500	7,876
# attacker veh.	9,671	Adj.	9,124	21,318	24,326	64	9 - 19	Unknown	50 - 500	119
Area (km^2)	3.6	1.61 - 155.95	1.61	6.51	1	8.7	0.11 - 1.47	0.0288	4.84	11.2
Beaconing rate (Hz)	1	Adj.	1	1	1	1	1 - 10	10	10 - 200	10 Hz
Possibility of packet collision	Extremely Low	Adj.	Extremely Low	Extremely Low	N/I	N/I	Extremely Low	Low	Low	Varies High
Shadowing model	Log Normal	Adj.	Simple Loss	Simple Path Loss	N/I	N/I	Friis Propagation Loss	Log Normal	Log Normal	Log Normal
RSSI support	Imp.	N/I	N/I	N/I	N/I	N/I	N/I	Imp.	Imp.	Imp.
Sybil attacks	N/I	Imp.	Imp.	Imp.	Imp.	N/I	N/I	Imp.	Imp.	Imp.
TX power control	N/I	N/I	N/I	N/I	N/I	N/I	N/I	N/I	Imp.	Imp.

Adj. denotes that the value/feature depends on simulator configuration or modifications on the traffic scenario which require development effort. Imp. and N/I denote implemented and not implemented, respectively

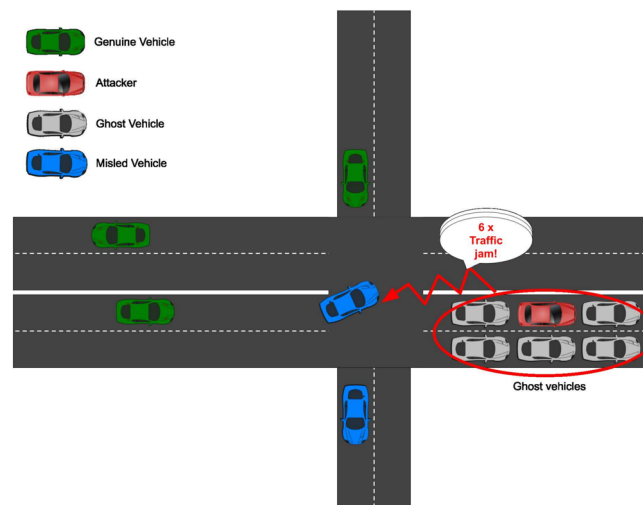


Fig. 1 An example sybil attack in which an attacker impersonates five more vehicles in order to disseminate a false traffic jam information

situations as well as regular traffic scenarios. Details of traffic generation in our dataset are given in Section 4.2.

3.3 Attacker model

The VANET environment relies on wireless communications, making it susceptible to attackers or malicious users who can easily gain control over the network. Consequently, an attacker can selectively sniff, drop, replay, reorder, and delay messages arbitrarily with negligible delay. We also assume that the attacker possesses a pool of valid pseudonym certificates, which can be obtained in several ways such as tampering with the storage device of an OBU or RSU describe in [22].

All messages transmitted within the network are signed. Therefore, while an attacker cannot directly modify the messages of other vehicles, their possession of a pool of valid certificates allows them to alter the signature and modify message fields as needed, or even create entirely new packets. Given that pseudonym identities are not linkable, receiving entities will not readily identify these packets as originating from an attacker. This anonymity makes it easy for the attacker to cause disruptions without being detected.

In this work, we focus on Sybil attacks in which an attacker uses multiple identities to spread false information. As a result, the other legitimate nodes will get fabricated information about the road condition, traffic flow, accidents or other ITS services. We also assume that attackers can modify their hardware to manipulate transmission power, an attack model exploiting this capability proposed by Yao et al. [21]. Several types of sybil attack methods have been described in the literature [5, 22]. These types are generated by manipulating two parameters. The first parameter is the method employed for generating message content, whereas the second is managing the transmission power.

In order to cover as many attack methods as possible, we included four data generation methods and five power control methods in our dataset. We formed 20 different attacks by combining data generation methods and power control schemes. Details of these combinations are given in Section 4.4

4 Simulation implementation details

Our simulation environment is built on top of the Veins [38] simulation tool. Developing and evaluating sybil attack detection methods require an environment close to the real world. Thus, we integrated various improvements. These improvements include correcting road structure, labeling surrounding structures, integrating appropriate radio propagation model, and employing urban traffic that reflects big cities. Components of the simulation system are given in Fig. 2.

During the implementation of the simulation environment, we employed a highly parametric approach. Simulation envi-

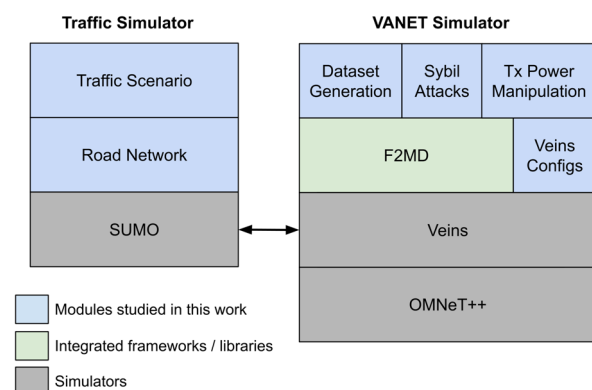


Fig. 2 Components of the proposed simulation system

ronment parameters include Veins settings, radio propagation model, sybil attack methods, maps, and vehicular traffic flows. Therefore, extending the dataset by running the simulations with different parameter values is always possible.

4.1 Map generation and modifications

We employed publicly available map data provided by OpenStreetMap (OSM) [39], which is a community project. In order to satisfy the requirements explained in Section 3, we performed several corrections and improvements on the map data. The city of Istanbul has a population of 15.46 million in an area of $5,343 \text{ km}^2$, according to 2020 data [40]. In order to reflect urban traffic conditions, we chose an area in Istanbul that includes neighborhoods from both Fatih and Beyoğlu districts. It has an area about 11 km^2 and its residential population is around 650,000 [41, 42]. The selected region also contains Bosphorus and The Golden Horn waterways. The region contains business traffic in the morning and evening hours, as it covers the Eurasia tunnel connecting the Asian and European continents and the Galata bridge connecting different districts. Moreover, the region covers highway scenarios due to bridges, light traffic scenarios at night, and dense traffic scenarios during the day. The region also includes several tourist attractions, including Hagia Sophia, Galata Tower, Grand Bazaar, Egyptian Bazaar, etc. Boundaries and the map of the region downloaded from OSM are given in Fig. 3.

A preliminary examination of the exported data revealed that the properties of many avenues, intersections, and streets are not defined according to the real world. This situation

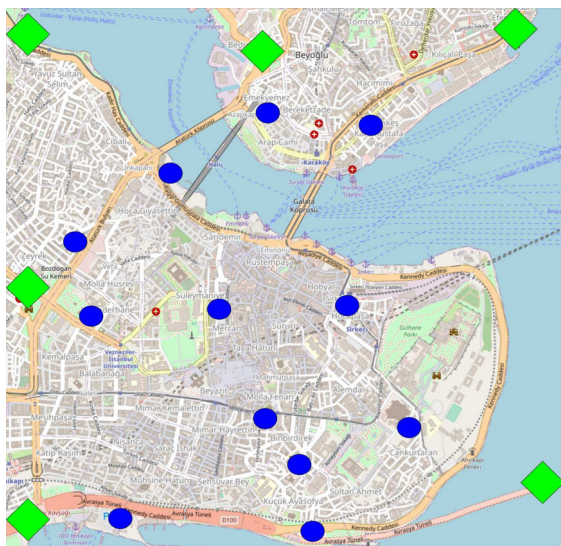


Fig. 3 Map of the region used in the simulation. Vehicles enter or leave the simulation area through six green squares representing outer region connections. Twelve parking lots are marked as blue circles. Both serve as sources and destinations for generated vehicle traffic

triggered us to inquire about other possible problems in the OSM data. Further investigation brings out different types of problems, such as incorrect number of lanes, undefined speed limits, missing connections in junctions and intersections, and misprogrammed traffic lights into the open. OSM data is created publicly with the contribution of its community. Thus correctness and freshness of OSM data highly depend on the number of contributors as well as their qualifications in an area. We believe these problems arise due to the limited number of contributors in Istanbul [43].

Correctness of the road network is the fundamental requirement of creating real-world like vehicle traffic in a simulation environment. Therefore, we conducted a detailed inquiry on OSM data and manually fixed all problems we found. The algorithm used in the map creation process is given Algorithm 1. Types and numbers of these fixes are given in Table 2.

Algorithm 1 Map creation process.

```

1: procedure MAP_CREATE
2:   Download map data via osmGet.py script [44]
3:   Generate SUMO compatible road network with osmBuild.py script [44]
4:   Manually check errors on SUMO road network
5:   if the road network contains OSM related errors then
6:     Fix errors on OSM website [39]
7:     Publish fixes on OSM website
8:     goto 2
9:   end if
10:  if the road network contains SUMO related errors then
11:    Fix errors using netedit tool
12:    goto 4
13:  end if
14:  Run basic traffic simulation on SUMO
15:  Manually check traffic flow for anomalies to detect road network errors
16:  if the road network contains errors then
17:    goto 4
18:  end if
19:  Export SUMO road network
20: end procedure

```

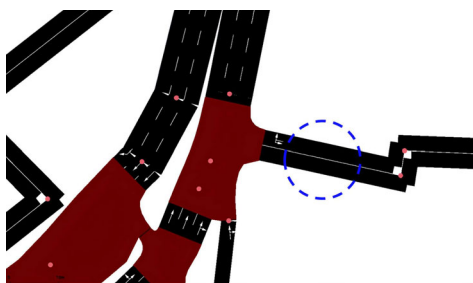
Corrections on the road network are made in three stages: online OSM data, offline OSM data, and SUMO data. Since the data provided by the OSM community is not sufficient for realistic simulations, the first correction step is data entry and modifications on OSM. While creating the road network with SUMO, errors usually occur due to the inability to detect some road relations by SUMO scripts. In order to solve such errors, some tricky operations, such as shifting road junctions, dividing road segments, and changing the number of lanes in road junctions, are required. The second stage includes these changes made only to local data, not published on OSM. In the last stage, the errors that cannot be corrected by modifying the OSM data are edited directly with the *netedit* [45] tool over the road network output of SUMO.

Table 2 Numbers and types of errors fixed manually on map data

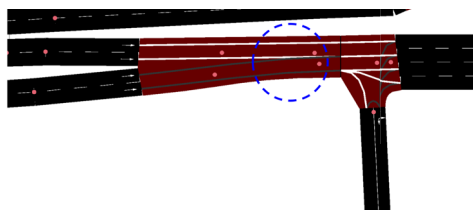
Type	Count
Incorrect number of lanes	113
Incorrect road type	350
Incorrect speed limit	341
Road junction errors	96
Incorrect turn restriction	107
Missing or incorrect traffic lights	25

One of the most important problems that we dealt with was the incorrectly defined number of lanes. Since it might dramatically increase or decrease the capacity of a road. This change in the road capacity affects the routing decisions of vehicles in the simulation environment and causes inaccurate traffic conditions.

Another significant problem that we revealed was the incorrect definition of road types. A minor road segment in OSM can be any of the residential road, living street, service road, and track road types. Most of the minor roads in the selected region were defined as residential roads, which allow bi-directional traffic flow at 50 kmph. However, as seen in Fig. 4b, some of these roads should be defined as living streets with lower speed limits.



(a) An incorrectly generated bidirectional road in SUMO



(c) An incorrectly generated junction in SUMO

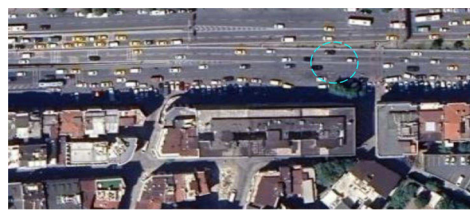
Other problems that cause inaccurate traffic flow include road junction errors and missing turn restrictions. Both of these problems result in routing decisions that are impossible in real life. An extreme case of such problems, which involves a main road connected to the wrong street, running over buildings, is given in Fig. 5.

The final problem we detected through analysis of the resulting traffic flow was the misprogrammed traffic lights. This group includes traffic lights that have any of the following problems. In most cases, some of the traffic lights in an intersection are not defined. In some cases, defined traffic lights have incorrect timing. In a few cases, defined traffic light does not control all lanes of the road segment.

In order to fix the aforementioned problems, we employed a manual approach. In cases of anomalies such as incorrectly defined number of lanes and road junction errors on the SUMO road network visible to the eye, we examined the exact location on Google Maps satellite images and then made necessary corrections. Whereas problems such as missing turn restrictions, incorrect definitions of road type, and misprogrammed traffic lights require a more time-consuming process. In such cases, we inspected the corresponding road segments along with their immediate neighbors through Google Street View. During the editing process, we examined over 5000 edges and intersections and manually edited



(b) Photograph of the road (a) from Google Street View [46]



(d) Satellite image of the road (c) from Google Maps [47]

Fig. 4 Examples of incorrect lane numbers road segments generated from OSM. An example of a street with an incorrect number of lanes is given in (a), where (b) is the photograph of the bidirectional street given in (a). Based on the photograph (b), it appears that two vehicles can not be able to cross this street simultaneously. (c) shows an example

where two paths are connected incorrectly, and (d) is the satellite image of (c) taken from Google Maps. The satellite image shows two two-lane roads are connected to form a four-lane road. In this example, both the number of lanes and the connections are incorrect



Fig. 5 An extreme example of road connection errors in OSM data

slightly over 1000 of them. Moreover, using the edge name patterns we automatically corrected over 8000 speed limit and road type errors.

After the correction process, we deleted some roads for optimization that we did not use in the simulation, such as pedestrian roads and internal roads of private facilities. Finally, the map was converted to SUMO format using the `osmBuild` script. This script also creates the routes of other transportation vehicles such as bus stops, train tracks.

The updated map, which includes corrections for the aforementioned errors, is available in SUMO format on GitHub¹.

4.2 Traffic generation

Traffic scenarios were created by manually defining the main flows to resemble the traffic information shared by the Istanbul Metropolitan Municipality (IMM) [48]. Three types of traffic flows that have been defined in our simulation. The first type is the traffic between outer regions. The second type is traffic between the outer and inner regions. The last type is randomly generated traffic that resembles the traffic flows in the inner region. We have determined 6 outer and 12 inner points that can be used as starting and ending regions for traffic flows. These locations are marked in Fig. 3. The outer points were selected based on the primary roads that link to the area, while the inner points were chosen around the parking lots in areas with high traffic. Then, we determined the traffic flow weights between the selected points using personal knowledge about the region and satellite images. After, we improved iteratively and experimentally by comparing it with the real-world traffic density map data.

¹ <https://github.com/VANET-Istanbul-Sybil-Attack-Dataset/dataset-src>

4.3 Veins configuration and modifications

It is necessary to define modules that affect radio propagation on the Veins settings to work with RSSI readings. The radio propagation model used in our study is based on the Log-Normal shadowing model used in VeReMi [12] and [11] study. Although F2MD [13] and VeReMi-Extension [14] studies have common features with VeReMi, they are unsuitable for the development of RSSI-based misbehavior detection algorithms since they utilize the simple path loss model as the radio propagation model.

Unlike previous studies [12, 14] that typically employed a 1 Hz beaconing rate, our simulation utilizes a 10 Hz rate. This tenfold increase aims to simulate a network experiencing significantly higher message density, reflecting real-world scenarios with heavy traffic volume. Importantly, this chosen value remains compliant with the allowed range within CAM standards (1-10 Hz) as outlined in [1, 31]. The specific Veins configuration details for this environment are provided in Table 3.

4.4 Attacker implementation

In our attacker model, we assume that attackers can capture packets over the wireless network, create new packets based on information from old packets, and broadcast new packets with negligible delay, enabling them to carry out replay attacks as described in [49]. We also assume that attackers can modify their hardware to manipulate transmission power, an attack model exploiting this capability proposed by Yao et al. [21]. Furthermore, we assume that the attackers can have a sufficient number of pseudonym certificates to perform a sybil attack, as mentioned in [22].

With the capabilities we described, attackers can monitor network activity to analyze existing messages, create new

Table 3 Veins configuration used in the simulation environment

Parameter	Value
Map area	3400m x 3300m
Vehicle count	1750 - 2250
Simulation time	4800 seconds
Beaconing rate	10 Hz
Obstacle shadowing	Simple Obstacle Shadowing with custom object parameters
Shadowing	Log-Normal
Attack txPower range	15mW - 25mw
Other NIC settings	Veins default settings - default txPower: 20mW - minPowerLevel: -110dBm - noiseFloor: -98dBm - antenna: monopole

Vehicle count reflects the number of active vehicles in the simulation after 10 minute initialization phase

packets and modify existing ones, sign packets with one of their pseudonyms, and broadcast with various transmission power control methods. Moreover, attackers can use multiple pseudonyms simultaneously and broadcast messages like multiple vehicles to perform a sybil attack. Attackers cannot use someone else's identity in generated packets. However, they can easily rebroadcast captured packets without modification using the original sender ID.

In our simulation environment, we used four data generation methods to perform sybil attacks:

- Attack 1: Generating attack packets completely random values
- Attack 2: Generating attack packets with random values within the range of nearby vehicles mobility information
- Attack 3: Reusing the values of old captured packets
- Attack 4: Generating attack packets with grid pattern for traffic jam scenario

In Attack 1, speed, acceleration, heading, and position data are generated randomly within speed limits and map size. In Attack 2, the attacker captures and inspects packets broadcast by nearby vehicles. It obtains the value ranges of other vehicles' position, speed, direction, and route data in the last 200 milliseconds time period. Then, it creates a new packet by generating random data within this value range. Both methods generate new packets independently of previously generated packets.

Attack 3 uses the third sybil attack generation method defined in [22]. The attacker constantly captures and inspects packets broadcast by nearby vehicles. It changes the timestamp and signature of the packets, signs with one of the attacker's pseudonyms, and broadcasts the packets. It is expected that attacks created in this way will be challenging to detect, as real data is used in this method.

In Attack 4, the attacker generates ghost vehicles in the grid pattern. In order to ensure that the vehicles are arranged in a grid form, the attacker's vehicle is considered the center, and ghost vehicles are placed around it. Position, speed, acceleration, and direction values of ghost vehicles are generated in such a way that there is an appropriate following distance between vehicles. As the attacker's vehicle moves, the surrounding vehicles continue to follow it with small variability.

In our simulation environment five output power control methods [11] can be used while performing sybil attacks:

- PC0: No power control
- PC1: Power control with random values
- PC2: Power control with rectangular pattern
- PC3: Power control with stairs pattern
- PC4: Power control with sawtooth pattern

Table 4 Vehicle types and distribution in our 25-minutes sample dataset

Vehicle type	Count
Non-attacker vehicles	7,876
Attacker vehicles	119
Ghost vehicles	652
Total number of physical vehicles	7,995

The ratio of the attacker (attacker probability) is determined in the simulation configuration, and its default value is 0.15. Vehicles selected as attackers decide the attack type when included in the simulation by choosing one attack type and one power control type. The number of ghost vehicles and attack-specific parameters are determined randomly.

4.5 Simulation execution and dataset generation

Our dataset was generated using Veins 5.2 and SUMO 1.8.0. The complete Veins project, source code, a 25-minute sample dataset, and accompanying documentation are available on GitHub². The simulation configuration is readily editable, allowing customization for specific research needs. To reproduce our dataset or generate new variations, users can simply rerun the supplied Veins project.

Our Veins project requires significant computational resources. On a modern desktop computer equipped with an AMD 5700X CPU and 64 GB RAM, generating a 25-minute dataset took approximately 20 days and resulted in a dataset of approximately 820 GB. Vehicle statistics are given in Table 4.

Table 5 compares our sample dataset with the VeReMi-Extension dataset [14] in terms of dataset size, simulation configuration, and output statistics. Our dataset simulates a significantly larger geographical area, roughly seven times the size of VeReMi-Extension. This expanded map facilitates more interactions among vehicles as they traverse longer distances. Furthermore, our dataset features higher vehicle density, enabling more interactions due to a message reaching a greater number of receivers. Additionally, our 10 Hz beaconing rate, a tenfold increase over VeReMi-Extension, significantly increases the number of messages generated per vehicle within the same period. Our simulation achieves a peak vehicle density of 164.73 vehicles/km², which is double the density observed in the VeReMi extension dataset. This increased density reflects a more intense traffic scenario, enabling us to evaluate system performance under congested conditions.

Our dataset output consists of log files for each vehicle and a shared ground truth file similar to VeReMi [12]. Vehicles

² <https://github.com/VANET-Istanbul-Sybil-Attack-Dataset/dataset-src>

Table 5 The comparison of our 25-minutes sample dataset with VeReMi Extension [14]

	Our dataset 25m scenario	VeReMi Extension [14] 24h scenario
Dataset size	820 GB	10.90 GB
Area	11.2 km ²	1.6 km ²
Simulation time	25 minute	24 hour
Beaconing rate	10 Hz	1 Hz
Peak density	164.73 Veh/km ²	67.4 Veh/km ²
Genuine vehicle	7,876	17,264
Ghost vehicles	652	7,399
Received genuine messages	849,231,132	14,742,569
Received attack messages	84,997,919	8,679,281
Transmitted genuine messages	21,112,206	1,821,920
Transmitted attack messages	2,003,527	1,228,581

record all messages they capture in the log file along with their own mobility information. Every attack and genuine message sent by vehicles is recorded in the ground truth file, along with the vehicle's mobility information and the actual values of the message. Vehicle log and ground truth files use the same data format. The log format is one line string, which consists of 4 main sections separated by "#" symbol, and the symbol "|" separates values within a section. Table 6 shows the detailed information of the data in the log records.

The section contains record type, which can be either 0 or 1, indicating ground truth or received captured message respectively. The second section contains the receiver's measurements, mobility information, and the received data for each message captured by the vehicle. The third section contains captured CAM contents. The last section consists of actual values of the CAM and attack information. The second section is only available in vehicle log files, whereas the fourth is only in the ground truth file.

There is a one-to-many relationship between ground truth and vehicle logs. Every message sent is recorded in the ground truth file. It is also recorded in the logs of the vehicles which the message reached. In order to perform studies such as classification or prediction on the dataset, records in the ground truth and vehicle log files can be matched via time, pseudonym, and message ID.

This dataset format aims to be used in algorithms to predict the sender's identity, mobility, attack type and power control. In addition, since the message is present in both the CAM and the Ground Truth sections, it is possible to use it in situations such as data corruption in packets due to hardware or software during data transmission.

5 Dataset validation

This section describes the validation process and presents results for our generated Sybil attack dataset. To demonstrate that our dataset fulfills the requirements of a realistic sybil

attack dataset we conducted a comprehensive assessment encompassing multiple aspects including traffic flow, RSSI measurements, attack distribution and packet collisions. In the following subsections we discuss the details of these validation processes.

5.1 Traffic flow

The data provided by IMM [48] is limited to the average speed of traffic flow on main roads; consequently, our validation process is restricted to these segments. To align with this approach, we adopted IMM's method of using the ratio of average speed to speed limit to calculate traffic density in our simulation. We then assessed the similarity between our simulated traffic density and real-life conditions using the normalized root mean square error (NRMSE), defined in formula Eq. 1. We calculate traffic density (y) using the ratio of average speed to speed limit. Within our analysis, y_i denotes the simulated traffic density for the i -th segment, while, $\hat{y}_i$ represents the corresponding traffic density in the IMM dataset, and n denotes the total road segment count.

$$NRMSE = \frac{\sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}}}{y_{max} - y_{min}} \quad (1)$$

As a result of comparing the IMM's traffic density data [48] between 10:00 and 11:00 on July 8, 2023, with our simulation output over 5-minute windows, the average NRMSE value was measured as 0.191.

5.2 RSSI measurement

Existing datasets like F2MD [13] and VeReMi Extension [14] rely on the simple path loss model, which is unable to capture real-world variations in signal strength. To address this limitation, we employed the log-normal shadowing model, which incorporates random variations in signal strength typically observed in real-world environments due to factors like

Table 6 Detailed explanation of data structure which is used in log files

Section	Data	Type
Type	Record Type	Integer
Receiver Info	RSSI	Float
	Self Time	Float
	Self Vehicle ID (real)	Integer
	Unused Padding	Integer (always 0)
	Self GPS Coordinates	Vector2D
	Self Position	Vector3D
	Self Position Confidence	Vector3D
	Self Speed	Vector3D
	Self Speed Confidence	Vector3D
	Self Heading	Vector3D
	Self Heading Confidence	Vector3D
	Self Acceleration	Vector3D
	Self Acceleration Confidence	Vector3D
	Self Vehicle Width	Float
	Self Vehicle Length	Float
Message	Time	Float
	Vehicle ID (pseudonym)	Integer
	Message ID	Integer
	GPS Coordinates	Vector2D
	Position	Vector3D
	Position Confidence	Vector3D
	Speed	Vector3D
	Speed Confidence	Vector3D
	Heading	Vector3D
	Heading Confidence	Vector3D
	Acceleration	Vector3D
	Acceleration Confidence	Vector3D
	Vehicle Width	Float
	Vehicle Length	Float
Ground Truth	Attack Type	Integer
	Power Control Type	Float
	Tx Power	Float
	Time	Float
	Vehicle ID (real)	Integer
	Message ID	Integer
	GPS Coordinates	Vector2D
	Position	Vector3D
	Position Confidence	Vector3D
	Speed	Vector3D
	Speed Confidence	Vector3D
	Heading	Vector3D
	Heading Confidence	Vector3D
	Acceleration	Vector3D
	Acceleration Confidence	Vector3D
	Vehicle Width	Float
	Vehicle Length	Float

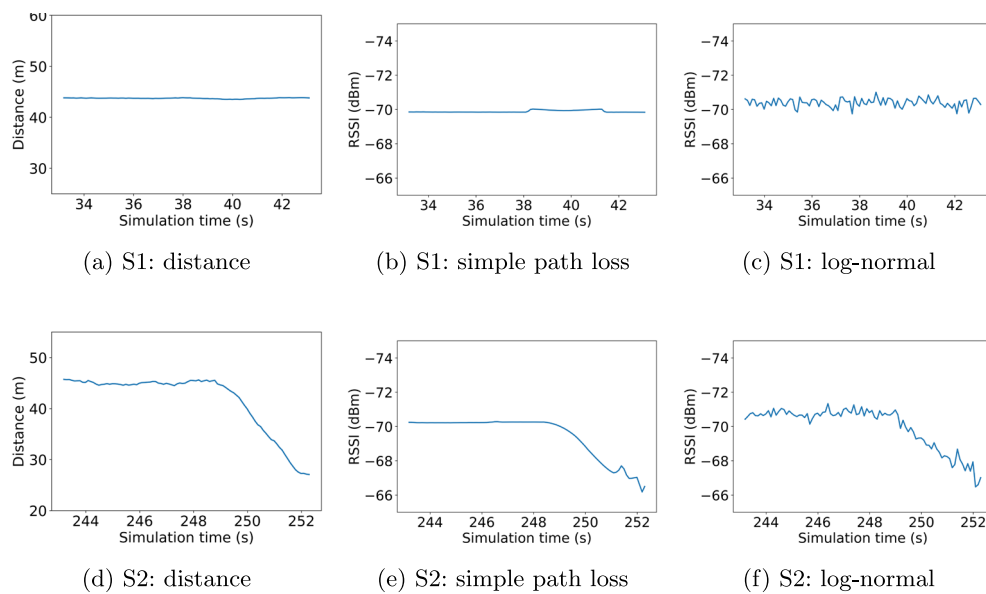


Fig. 6 Comparison of RSSI sequences generated by F2MD [13] and our simulation, demonstrating differences in signal strength calculation over distance. This figure presents a comparative visualization of two scenarios, each simulated twice using F2MD's configuration and our simulation environment with refined signal propagation configuration.

multipath propagation and obstacles. Figure 6 demonstrates a comparison of RSSI sequences generated by simple path loss and log-normal shadowing. To ensure a direct comparison, identical traffic scenarios were used for both simulations. As expected, both simulations exhibit an inverse proportionality between signal strength and distance. However, log-normal shadowing produces more realistic RSSI readings.

5.3 Attack distribution

Developing and testing sybil attack detection methods requires a dataset with a substantial number of interactions between sybil attackers and genuine vehicles. These interac-

The simulations capture identical moments for a direct comparison of RSSI calculation methods. A close examination of the RSSI sequences across subfigures (b, c) and (e, f) suggests that our simulation produces more realistic variations in signal strength over distance

tions should exhibit diverse durations to reflect the variability of real-world scenarios.

Our simulation includes attacker vehicles, making up 1.5% of a total of 7876 vehicles. Each attacker randomly creates between 2 and 10 ghost vehicles. Table 4 shows that the number of attacker vehicles and ghost vehicles resides in these boundaries.

We further evaluate the validity of the interactions between ghost vehicles and genuine vehicles. Examining logs of received messages belonging to genuine vehicles, we found that there were 314,592 interactions. Figure 7 visualizes the distribution of these interactions, highlighting the number of interactions that occurred based on the number of messages exchanged between ghost and genuine vehicles.

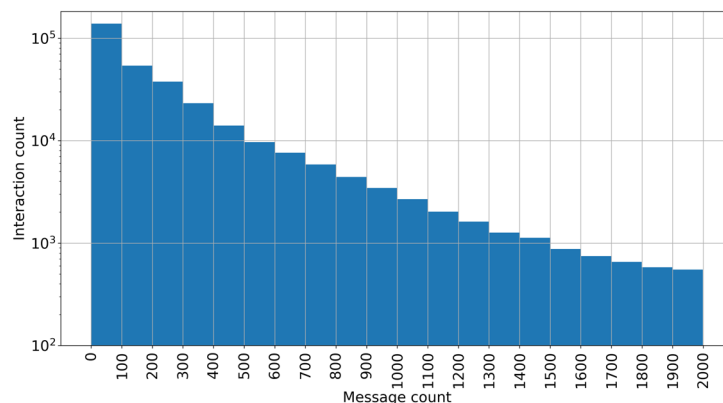


Fig. 7 Distribution of ghost-genuine vehicle interactions

5.4 Packet collision

Existing datasets [12, 14] employ low traffic densities, resulting in significantly lower probabilities of collision. In the real world, we experience traffic jams causing higher vehicle densities. This leads to channel capacity being exceeded, resulting in packet losses. Higher vehicle densities also increase the probability of signal interference, further contributing to packet losses. In order to validate that our dataset reflects these situations, we inspected several statistics available in the Veins environment.

Veins [26] reports packet statistics for the entire simulation. To enable comparison of different traffic intensities, we conducted independent simulations of both LuST [14] and our traffic scenario for two different time ranges. For LuST, we selected 8:00 AM (LuST-A) and 7:00 PM (LuST-B). Within our traffic scenario, we chose the 5th minute (IST-A) and 25th minute (IST-B) to reflect contrasting traffic conditions. Figure 8 shows histograms about this statistics by comparing our simulation results against the VeReMi Extensions LuST traffic scenario [14] for metrics received message count, collision count, and signal-to-noise and interference

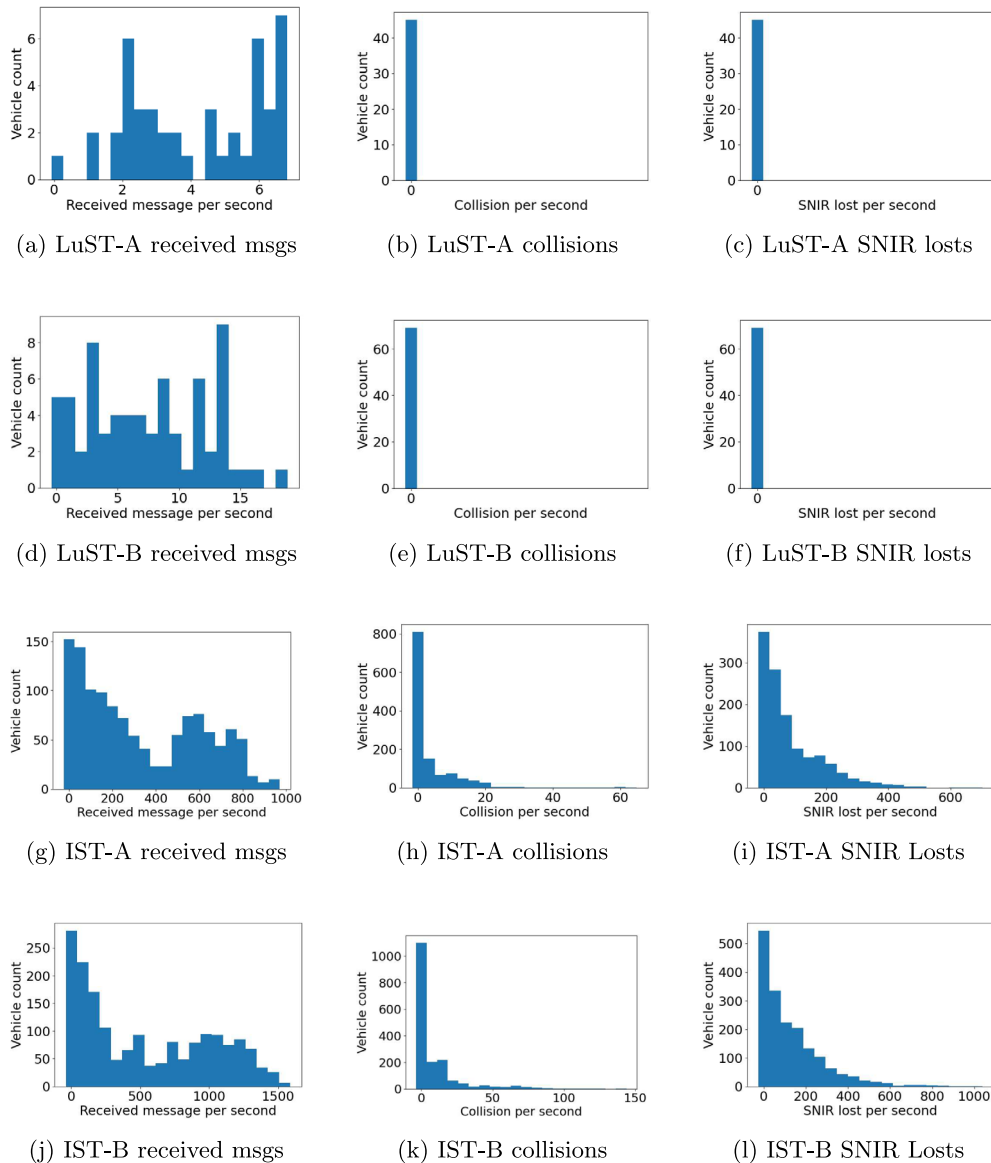


Fig. 8 Comparison of received message counts and packet loss statistics between our simulation and the VeReMi Extension LuST scenario [14]. This figure includes four independent tests: two different time

ranges in our Istanbul scenario (IST-A: 5th minute for average traffic, IST-B: 25th minute for high traffic) and two time ranges within the LuST scenario (LuST-A: 8:00 AM and LuST-B: 7:00 PM)

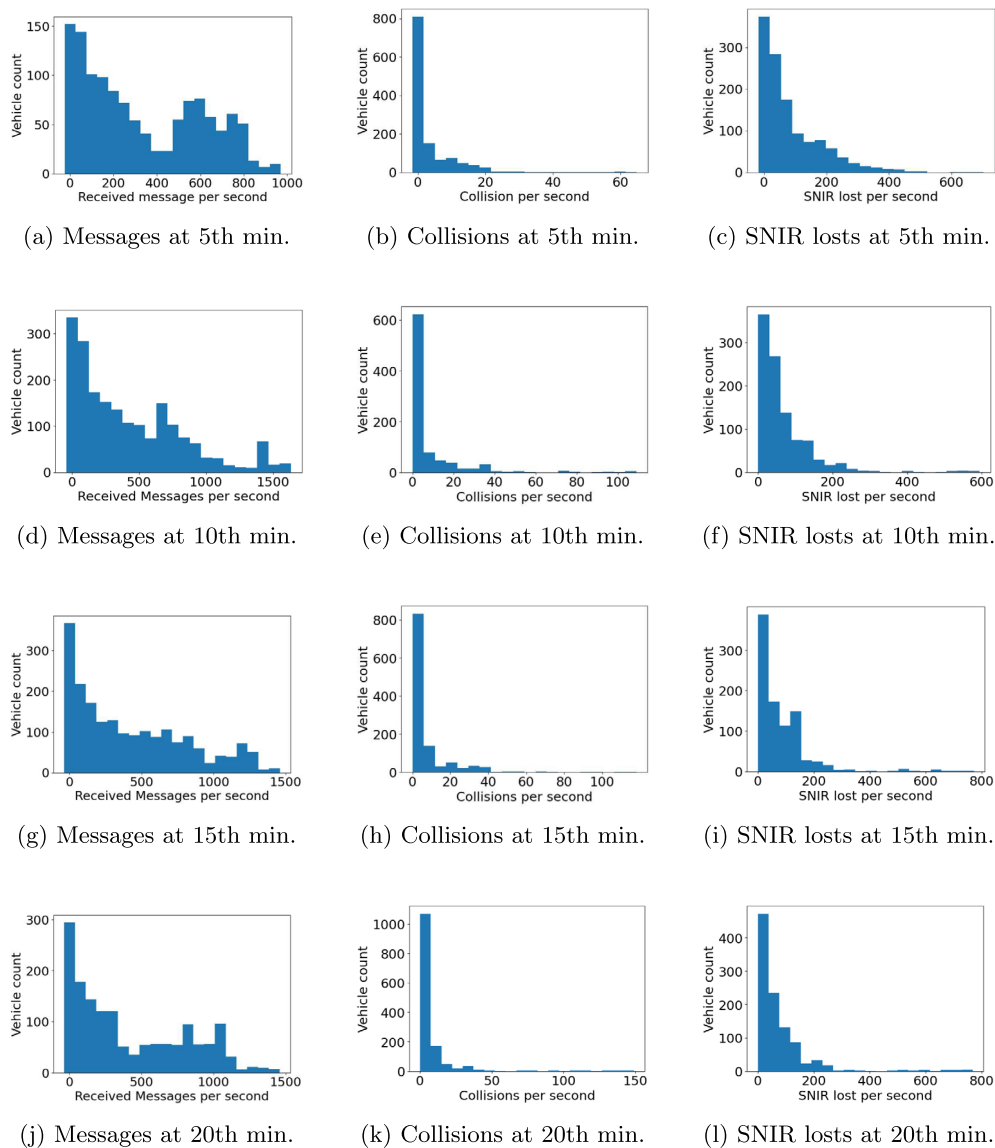


Fig. 9 Received message counts and packet loss statistics for 5th, 10th, 15th and 20th minutes of our Istanbul scenario. The statistics were calculated independently for each distinct minute

ratio (SNIR) lost count. The collision count metric quantifies packet loss due to bit errors resulting from collisions, while the SNIR lost count metric measures packet loss caused by insufficient signal power. By looking at both received message counts and collisions, the analysis suggests the proposed scenario has a significantly higher traffic density compared to the VeReMi Extensions LuST scenario. It's important to note that the VeReMi Extensions LuST traffic scenario specifically produces zero collisions. To further facilitate a detailed examination of our simulation's temporal dynamics, Fig. 9 presents histograms illustrating the received message count, collision count, and SNIR lost count metrics at the 5th, 10th, 15th, and 20th minutes of our Istanbul scenario.

6 Conclusion

In this work, we created a simulation environment tailored for developing and testing Sybil attack detection methods in VANETs, with the key objective of generating a dataset suitable for RSSI-based approaches. This dataset incorporates various Sybil attack strategies, including existing power control-based attacks, to enable the development of robust detection mechanisms that are resilient to such attacks. To the best of our knowledge, no existing public dataset offers these combined features. We imported a real-life Istanbul road map covering approximately 11 km^2 and, after manual error correction, designed a realistic, high-density traffic

scenario based on IMM data. We carefully calibrated Veins' radio propagation model and parameters to ensure accurate RSSI readings. New Sybil attack and power control methods were integrated, and a versatile dataset format was defined. Critically, we have made our simulation source code, settings, scenarios, and a sample output publicly available³.

Our work highlights the importance of incorporating diverse building and landscape data into simulations for more accurate signal propagation modeling. Future research should investigate the precise effects of these elements on VANET signals to create truly realistic simulation environments. Specifically, determining the parameters needed to integrate these signal effects accurately into Veins would be a valuable contribution. To fully utilize this enhanced modeling, a collaborative effort to manually label buildings and landscapes on public map services like OpenStreetMap with detailed information (e.g., building type, construction material) would be highly beneficial.

While our simulation environment represents a significant step towards realism, it also has limitations. The high computational cost of realistic, high-traffic-density VANET simulations is a major challenge. Our full simulation scenario currently takes approximately 55–60 days to execute on a modern computer. This limitation stems from Veins' lack of inherent support for multi-threaded or multi-process simulation. Enabling parallel execution of simulations would significantly accelerate VANET simulation research. We are actively investigating methods for parallelizing our simulation process, which we anticipate will lead to substantial reductions in simulation time.

Furthermore, while we have integrated power control into our Sybil attacks, these attacks are currently relatively primitive in their use of environmental information and attack parameter strategizing. Future work will focus on developing more sophisticated and realistic attack models that leverage environmental context and employ more advanced attack strategies. This will involve examining existing attack detection methods in detail and exploring more complex attacks that pose real-world risks. We also plan to investigate machine learning-based detection methods that incorporate RSSI readings to detect these more complex Sybil attacks. Finally, although our dataset incorporates realistic traffic patterns and RSSI characteristics, further validation against real-world measurements would strengthen its credibility and utility. Gathering and analyzing real-world RSSI data in diverse traffic conditions would be a valuable next step.

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³ <https://github.com/VANET-Istanbul-Sybil-Attack-Dataset/dataset-src>

Author Contributions -Study conception and design (Z.C.T.)
-Implementation and data acquisition (T.G., Z.C.T.)
-Analysis and interpretation of data (T.G., Z.C.T.)
-Drafting of manuscript (T.G., Z.C.T.)
-Critical revision (Z.C.T.)

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Data Availability The dataset and source code are provided on the GitHub repository. This repository is accessible from the URL <https://github.com/VANET-Istanbul-Sybil-Attack-Dataset/dataset-src>.

Code Availability The source code used in this study is provided in the same repository with the dataset.

Declarations

Competing Interests The authors declare no competing interests.

Ethical Approval Not applicable.

Human and Animal Rights This article does not contain any studies on human or animal subjects.

Informed Consent Not applicable.

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